2 A1 Use of alpha emitter as source (one mark), with a reason (strongly ionising) (one mark). These are independent marks. **[2]**
Do not award these marks if the source is not sensible (e.g. smoke producing alpha particles)

A2 Correct circuit showing milliammeter/microammeter/galvanometer **[1]**
Do not allow ammeter. Accept 'sensitive ammeter'.
A power supply must be shown.

A3 Use of Bourdon gauge/pressure gauge/manometer/barometer **[1]**
Accept pressure sensor + datalogger.
Do not allow vague 'pressure metre'.
Could be shown on the diagram.

B1 Change air pressure using pump (could be shown on diagram) **[1]**

B2 Diagram showing 'closed box' surrounding the equipment. **[1]**
do not award this mark if the gauge is not measuring the pressure in the box.

B3 Procedure **[1]**
Measure ionisation current and *P*; change air pressure (and measure new pressure and ionisation current).
This mark can be scored even if the design is unworkable.
Accept a table.

C1/C2 Two safety precautions **[2]**
e.g. use source handling tool
store source in lead lined box when not in use
do not point source at people/do not look directly at source
use safety goggles when dealing with low/high pressures
Do not allow vague 'safety goggles' without clarification
container must be strong enough to withstand high/low pressure
use safety screens in case of implosion/explosion
Do not allow lead suits/lead lined rooms/lead gloves/lead lined expt. etc.
One mark for safety relating to radiation.
One mark for safety relating to pressure.

D Any good/further detail **[1]**
Examples of creditworthy points might be:
Use high voltage as current is small
Use GM tube and scalar/ratemeter to monitor activity to ensure that it does not change
Use of source of long half-life
Place source close to plates (as range of alpha in air is small)
Tap gauge when taking readings (in case needle sticks)
Fix position of source relative to plates
Separation of plates constant
Allow other valid points.

[Total: 10 marks]

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